

JESS Thermodynamic Database v8.9

Xenon

24-Jan-23

Reaction No. 70235							
$\text{XeO}_3 + \text{OH-1} = \text{H+1_XeO}_4\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.8(2SF)	4	ENS	7276

Reaction No. 70236							
$\text{H+1_XeO}_6\text{-4} + \text{OH-1} = \text{XeO}_6\text{-4} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.47(2SF)	4	ENS	7276

Reaction No. 70237							
$\text{H+1(2)_XeO}_6\text{-4} + \text{OH-1} = \text{H+1_XeO}_6\text{-4} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(2SF)	4	ENS	7276

Reaction No. 70238							
$\text{H+1(4)_XeO}_6\text{-4} + 2\text{<H+1>} + 2\text{<e-1>} = \text{XeO}_3 + 3\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.36(3SF)	4	ENS	7276

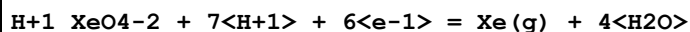
Reaction No. 70239							
$\text{XeO}_3 + 6\text{<H+1>} + 6\text{<e-1>} = \text{Xe(g)} + 3\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.12(3SF)	0	ENS	7276

Reaction No. 70240							
$\text{XeF}_2 + 2\text{<e-1>} = \text{Xe(g)} + 2\text{<F-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.64(3SF)	4	ENS	7276

Reaction No. 70241							
$\text{H+1_XeO}_6\text{-4} + 4\text{<H+1>} + 2\text{<e-1>} = \text{H+1_XeO}_4\text{-2} + 2\text{<H}_2\text{O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	E:0.94 (2SF)	4	ENS	7276
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Reaction No. 70242



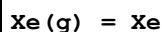
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.26 (3SF)	0	ENS	7276

Reaction No. 70244



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:105. (3SF) kJ	4	ENS	7276

Reaction No. 72417



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:3.22 (3SF)	4	CRV	10174
2	50	0	Inf. Dilution	dG:2.80 (3SF)	3	CRV	10174
3	75	0	Inf. Dilution	dG:2.26 (3SF)	3	CRV	10174
4	100	0	Inf. Dilution	dG:1.63 (3SF)	3	CRV	10174
5	125	0	Inf. Dilution	dG:0.92 (2SF)	2	CRV	10174
6	150	0	Inf. Dilution	dG:0.13 (2SF)	2	CRV	10174
7	175	0	Inf. Dilution	dG:-0.73 (2SF)	2	CRV	10174
8	200	0	Inf. Dilution	dG:-1.66 (3SF)	1	CRV	10174
9	225	0	Inf. Dilution	dG:-2.65 (3SF)	1	CRV	10174
10	250	0	Inf. Dilution	dG:-3.71 (3SF)	1	CRV	10174
11	300	0	Inf. Dilution	dG:-6.04 (3SF)	0	CRV	10174

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination

4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
ENS	Not specified

References

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 10174(95OHS)

J. Phys. Chem. Ref. Data, 1995, 24, 1401 - 1560; Oelkers EH; Helgeson HC; Shock EL; Sverjensky DA; Johnson JW; Pokrovskii VA; Summary of the Apparent Standard Partial Molal Gibbs Free Energies of Formation of Aqueous Species, Minerals, and Gases at Pressures 1 to 5000 Bars and Temperatures 25 to 1000C